

Relation Algebra

§1. Notation

Traditionally Adjunction $F \dashv G$ is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

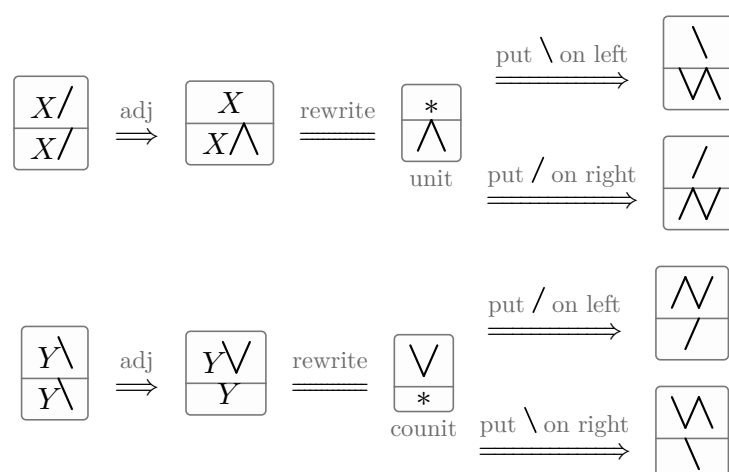
By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

(1a)

- write F as $/$, G as $\backslash$
- render $a \leq b$ as $\boxed{\frac{a}{b}}$
- **rewrite** $X \leq XF$ as $1 \leq F$

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}}$$

adj



now we have $\wedge\wedge = /$ and $\wedge\backslash = \backslash$, and you just discovered string diagrams.

§1.1. Adjunctions

(1.1a)

$F \dashv G$	F 's type	monad FG	$1 \sqsubseteq FG$	$GF \sqsubseteq 1$	F preserves u	G preserves n	$FGF=F$	$GFG=G$
$\triangleleft \dashv \triangleleft$	$A \rightarrow A \otimes A$	$\triangleleft = 1$	$1 \sqsubseteq \triangleleft$	$\triangleleft \sqsubseteq 1$	$(R \cup S) \triangleleft = R \triangleleft \cup S \triangleleft$	$(R \cap S) \triangleright = R \triangleright \cap S \triangleright$	$\triangleleft \triangleleft = \triangleleft$	$\triangleright \triangleright = \triangleright$
$\multimap \dashv \multimap$	$A \rightarrow I$	$\multimap = T$	$1 \sqsubseteq \multimap$	$\multimap \sqsubseteq 1$	$(R \cup S) \multimap = R \multimap \cup S \multimap$	$(R \cap S) \multimap = R \multimap \cap S \multimap$	$\multimap \multimap = \multimap$	$\multimap \multimap = \multimap$
$\circ \dashv \circ$	$(A \rightarrow B) \rightarrow (B \rightarrow A)$	$R^\circ \circ = R$	$R^\circ \circ = R$	$R^\circ \circ = R$	$(R \cup S)^\circ = S^\circ \cup R^\circ$	$(R \cap S)^\circ = R^\circ \cap S^\circ$	$R^\circ \circ = R$	$R^\circ \circ = R$
$\circ \dashv \triangleleft \dashv \multimap$	$(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$	1	$(\circ \triangleleft \otimes 1) = 1$ $(1 \otimes \triangleright \multimap) = 1$	$(1 \otimes \circ \triangleleft) = 1$ $(\triangleright \multimap \otimes 1) = 1$	$=$	$=$	$=$	$=$
$\triangleright \dashv \multimap \dashv \triangleleft$	$(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$	1	$(1 \otimes \circ \triangleleft) = 1$ $(\triangleright \multimap \otimes 1) = 1$	$(\circ \triangleleft \otimes 1) = 1$ $(1 \otimes \triangleright \multimap) = 1$	$=$	$=$	$=$	$=$
$\Delta \dashv \cap$	$(A \rightarrow B) \rightarrow (A \rightarrow B)^2$	$R \cap R = R$	$R \cap R = R$	$R \cap S \sqsubseteq R$	$\Delta(R \cup S) = \Delta R \cup \Delta S$	$(R \cap T) \cap (S \cap U) = (R \cap S) \cap (T \cap U)$	$R \cap R = R$	$R \cap R = R$
$u \dashv \Delta$	$(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$	$(R, S) \mapsto (R \cup S, R \cap S)$	$R \sqsubseteq R \cup S$	$R \cup R = R$	$(R \cup T) \cup (S \cup U) = (R \cup S) \cup (T \cup U)$	$\Delta(R \cap S) = \Delta R \cap \Delta S$	$R \cup R = R$	$R \cup R = R$
$\emptyset \dashv !$	$\{*\} \rightarrow (A \rightarrow B)$			$\emptyset \sqsubseteq R$				
$\cdot S \dashv / S$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	S/S	$R \sqsubseteq (RS)/S$	$(T/S) S \sqsubseteq T$	$(S \cup T)R = SR \cup TR$	$(R \cap R')/S = (R/S) \cap (R'/S)$	$((RS)/S)S = RS$	$((T/S)S)/S = T/S$
$S \cdot \dashv S \setminus$	$(B \rightarrow C) \rightarrow (A \rightarrow C)$	$S \setminus S$	$R \sqsubseteq S \setminus (SR)$	$S(S \setminus T) \sqsubseteq T$	$R(S \cup T) = RS \cup RT$	$S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$	$S(S \setminus (SR)) = SR$	$S \setminus (S(S \setminus T)) = S \setminus T$
$R \cap \dashv R \Rightarrow$	$(A \rightarrow B) \rightarrow (A \rightarrow B)$	$X \mapsto R \Rightarrow (X \cap R)$	$X \sqsubseteq R \Rightarrow (X \cap R)$	$R \cap (R \Rightarrow Y) \sqsubseteq Y$	$R \cap (X \cup Y) = (R \cap X) \cup (R \cap Y)$	$R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$	$R \cap (R \Rightarrow (X \cap R)) = X \cap R$	$R \Rightarrow (R \cap (R \Rightarrow Y)) = R \Rightarrow Y$
$\mathcal{D} \dashv \cdot T$	$(A \rightarrow B) \rightarrow \text{Cor } A$	$R \mapsto (\mathcal{D}R)T$	$R \sqsubseteq (\mathcal{D}R)T$	$\mathcal{D}(AT) \sqsubseteq A$	$\mathcal{D}(R \cup S) = \mathcal{D}R \cup \mathcal{D}S$	$(A \cap B)T = AT \cap BT$	$\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$	$(\mathcal{D}(AT))T = AT$
$\mathcal{R} \dashv T \cdot$	$(A \rightarrow B) \rightarrow \text{Cor } B$	$R \mapsto T(\mathcal{R}R)$	$R \sqsubseteq T(\mathcal{R}R)$	$\mathcal{R}(TA) \sqsubseteq A$	$\mathcal{R}(R \cup S) = \mathcal{R}R \cup \mathcal{R}S$	$T(A \cap B) = TA \cap TB$	$\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$	$T(\mathcal{R}(TA)) = TA$
$\cdot f \dashv \cdot f^\circ$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \sqsubseteq ff^\circ$	$f^\circ f \sqsubseteq 1$	$(S \cup T)R = SR \cup TR$	$\text{Simple}(g) \Rightarrow (X \cap Y)g^\circ = Xg^\circ \cap Yg^\circ$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$f^\circ \dashv \cdot f \cdot$	$(A \rightarrow C) \rightarrow (B \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \sqsubseteq ff^\circ$	$f^\circ f \sqsubseteq 1$	$R(S \cup T) = RS \cup RT$	$f(X \cap Y) = fX \cap fY$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$i \dashv E$	$\text{Map} \mapsto \text{Rel}$	E	$\frac{1}{\exists} : A \rightarrow EA$	$\exists : EB \rightarrow B$	—	—	$\frac{\exists}{\exists} = 1$	$\frac{R}{\exists} \exists = R$
$\cdot \exists \dashv \frac{\exists}{\exists}$	$\text{Map}(A, EB) \rightarrow (A \rightarrow B)$	1	$\frac{f\exists}{\exists} = f$	$\frac{R}{\exists} \exists = R$	—	—	$\frac{\frac{R}{\exists} \exists}{\exists} = \frac{R}{\exists}$	$\frac{\frac{R}{\exists} \exists}{\exists} = \frac{R}{\exists}$
$\frac{\exists}{\exists} \dashv \cdot \exists$	$(A \rightarrow B) \rightarrow \text{Map}(A, EB)$	1	$\frac{R}{\exists} \exists = R$	$\frac{f\exists}{\exists} = f$	—	—	$\frac{\frac{R}{\exists} \exists}{\exists} = \frac{R}{\exists}$	$\frac{\frac{f\exists}{\exists} \exists}{\exists} = \frac{f\exists}{\exists}$

§1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
 \begin{array}{ccccc}
 \boxed{\begin{array}{c} X \\ f(R/T) \end{array}} & \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} & \boxed{\begin{array}{c} f^\circ X \\ R/T \end{array}} & \xLeftrightarrow{\cdot T \dashv / T} & \boxed{\begin{array}{c} (f^\circ X)T \\ R \end{array}} \\
 \text{f a map} & & & & \\
 \xLeftrightarrow{\text{associativity}} & \boxed{\begin{array}{c} f^\circ(XT) \\ R \end{array}} & \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} & \boxed{\begin{array}{c} XT \\ fR \end{array}} & \xLeftrightarrow{\cdot T \dashv / T} & \boxed{\begin{array}{c} X \\ (fR)/T \end{array}} \\
 \xRightarrow{\text{indirect equality}} & & & & \boxed{\begin{array}{c} f(R/T) \\ (fR)/T \end{array}}
 \end{array}$$

$$\text{Map}(f) \Rightarrow f(R/S) = fR/S$$

(1.2b)

	$\cdot T$	$T \cdot$	$\cdot g$	$g^\circ \cdot$	Tn	$^\circ$	$\leftarrow \triangleleft$	Δ
$\cdot S$	$R/S_1 \ S_2=R/S_2/S_1$	$S \setminus (R/W) = (S \setminus R)/W$	$R/(Sg) = (Rg^\circ)/S$	$\text{Map}(f) \implies f(R/S) = fR/S$	—	$(Y/S)^\circ = S^\circ \setminus (Y^\circ)$	$(RS)^\wedge = R^\wedge (S \otimes 1)$	$(RnR')/S = (R/S) \cap (R'/S)$
$S \cdot$	$S \setminus (R/W) = (S \setminus R)/W$	$S_1 \ S_2 \setminus R = S_2 \setminus (S_1 \setminus R)$	$S \setminus (Rg^\circ) = (S \setminus R)g^\circ$	$(g^\circ S) \setminus R = S \setminus (gR)$	—	$(S \setminus Y)^\circ = Y^\circ / S^\circ$	$((S \otimes 1)R)^\wedge = SR^\wedge$	$S \setminus (RnR') = (S \setminus R) \cap (S \setminus R')$
$\cdot f$	$\text{Map}(f) \implies R/fS = (R/S)f^\circ$	$T \setminus (Rf^\circ) = (T \setminus R)f^\circ$	$(RS)^\circ = S^\circ R^\circ$			$(RS)^\circ = S^\circ R^\circ$		$\text{Simple}(g) \implies (XnY)g^\circ = Xg^\circ nYg^\circ$
$f^\circ \cdot$	$\text{Map}(f) \implies f(R/S) = fR/S$	$(Tf^\circ) \setminus R = f(T \setminus R)$	—	$(RS)^\circ = S^\circ R^\circ$		$(RS)^\circ = S^\circ R^\circ$		$f(T_1 n T_2) = fT_1 n fT_2$
Rn	—	—	—	—	$R \Rightarrow S \Rightarrow T = RnS \Rightarrow T$	$(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$	—	$R \Rightarrow SnT = (R \Rightarrow S) \cap (R \Rightarrow T)$
$^\circ$	$(Y/T)^\circ = T^\circ \setminus (Y^\circ)$	$(T \setminus Y)^\circ = Y^\circ / T^\circ$	$(RS)^\circ = S^\circ R^\circ$	$(RS)^\circ = S^\circ R^\circ$	$(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$	$R^\circ \circ = R$	$(R^\wedge)^\circ = (R^\circ \otimes 1)$ $(1 \otimes \triangleright \rightarrow)$ both bends: the $^\circ$ section's display	$(T_1 n T_2)^\circ = T_1^\circ n T_2^\circ$
$\leftarrow \triangleleft$	$(RT)^\wedge = R^\wedge (T \otimes 1)$	$((T \otimes 1)R)^\wedge = TR^\wedge$	—	—	—	$(R^\wedge)^\circ = (R^\circ \otimes 1)$ $(1 \otimes \triangleright \rightarrow)$ both bends: the $^\circ$ section's display	—	—
u	$(S \cup T)R = SR \cup TR$	$R(S \cup T) = RS \cup RT$	$(S \cup T)R = SR \cup TR$	$R(S \cup T) = RS \cup RT$	$Rn(S \cup T) = RnS \cup RnT$	$(R \cup S)^\circ = S^\circ \cup R^\circ$		
$\emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$Rn \emptyset = \emptyset$	$\emptyset^\circ = \emptyset$		

§1.3. Adjoint triples

Beyond $u \dashv \Delta \dashv n$, which §1.1 already carries as two rows, the table holds one adjoint triple with content: $\exists_f \dashv f^* \dashv \forall_f$ along a map f , once on each side of the composite. $f : A \rightarrow B$ throughout this subsection.

(1.3a)

$$\begin{aligned} \cdot f \dashv \cdot f^\circ \dashv /f^\circ \\ f^\circ \dashv \cdot f \dashv \cdot f \setminus \end{aligned}$$

The first two links of each chain are the map shunting rules, and the third is §1.1's division row read at a chosen divisor: $\cdot S \dashv /S$ at $S := f^\circ$, and $S \cdot \dashv S \setminus$ at $S := f$.

The third link is not the first over again. For a map f , $/f$ collapses to $\cdot f^\circ$; being a map is exactly what that collapse spends $\text{Map}(f) \implies R/fS = (R/S)f^\circ$ — f° is not a map, so $/f^\circ$ does not collapse in turn, and the three operators stay distinct. On $\text{Rel}(\mathbb{I}, A) = \text{EA}$ they are the image triple:

(1.3b)

operator	acts by	name
$\cdot f$	$S \mapsto f[S]$	direct image, $\exists_f$
$\cdot f^\circ$	$T \mapsto f^{-1}[T]$	inverse image, f^*
$/f^\circ$	$S \mapsto \{b : f^{-1}(b) \subseteq S\}$	$\forall_f$

§1.1's $\mathcal{D} \dashv \cdot T$ is this same chain along the projection $A \otimes B \rightarrow A$, read through $\text{Rel}(A, B) = E(A \otimes B)$: $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$, AT is the relation that ignores b altogether, and the third link is $R \mapsto 1 \cap R / T = \{(a, a) : \forall b. aRb\}$.

(1.3c)

$$\mathcal{D} \dashv \cdot T \dashv 1 \cap \cdot / T$$

Three is where it stops, and one map breaks both ends. Take $A = \{a_1, a_2\}$ and $B = \{b\}$: $f[\{a_1\}] \cap f[\{a_2\}] = \{b\}$ while $f[\{a_1\} \cap \{a_2\}] = \emptyset$, so $\cdot f$ does not preserve meets and has no left adjoint, and the same two subsets give $\forall_f(\{a_1\} \cup \{a_2\}) = \{b\}$ against $\forall_f\{a_1\} \cup \forall_f\{a_2\} = \emptyset$, so $/f^\circ$ does not preserve joins and has no right adjoint. A monic f restores the binary case and no more — Tf still reaches only $\text{im } f$, and $\forall_f f \emptyset = B \setminus \text{im } f$ is still not $\emptyset$. The chain extends only when f° is a map as well, and then $/f^\circ = \cdot f^\circ = \cdot f$ and it repeats forever. For an S with neither S nor S° a map there is no triple at all: $\cdot S \dashv /S$ is two links and stops.

(1.2b)

The rows that chain forever say nothing by it: $\circ$ is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and $\circ \dashv \circ \dashv \circ$ has no end, and §1.1's other row of that kind, $\circ \triangleleft \dashv \triangleright \circ$, goes the same way.

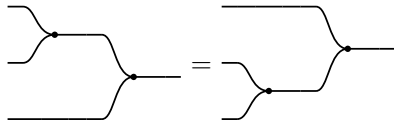
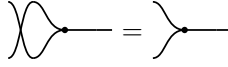
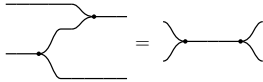
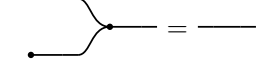
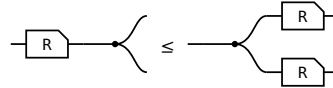
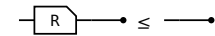
§2. Relations

(2a)

$\mathbf{Rel}$ is a poset-enriched category with $(\otimes, \circ)$ where $\otimes$ is commutative cartesian product, and converse $\circ^{-1}$, and

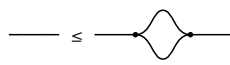
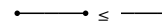
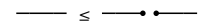
$C: (\triangleright : A \otimes A \rightarrow A, \circ : I \rightarrow A)$ a commutative monoid with $C^\circ \dashv C$. We write C° as $(\triangleleft : A \rightarrow A \otimes A, \circ : A \rightarrow I)$

(2b)

 $\triangleright$ associative $\triangleright$ commutative $\circ$ is its unitFrobenius, one half — the other is its $\circ$ $R \triangleleft \leq \triangleleft (R \otimes R)$ $R \circ \leq \circ$

§2.1. $\forall$ object A , $(A, \triangleleft, \circ) \dashv (A, \triangleright, \circ)$

(2.1a)

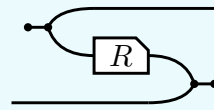
 $\triangleleft \leq 1$  $1 \leq \triangleleft$  $\circ \leq 1$  $1 \leq \circ$

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In $\mathbf{Rel}$ it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§3. $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

(3a)

$\circ$ is primitive, part of the data the first section lists. It turns both of R 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R° , not its definition:



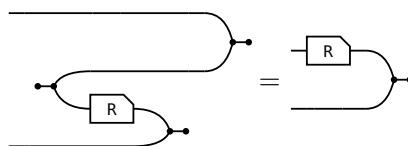
$$R^\circ = (\circ \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \multimap \circ)$$

where $\circ \triangleleft : \mathbb{I} \rightarrow \mathbf{a} \otimes \mathbf{a}$ opens a pair of wires out of nothing and $\triangleright \circ : \mathbf{b} \otimes \mathbf{b} \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\multimap$ are $\triangleright^\circ$ and $\circ^\circ$. The laws of $\circ$ are that it is a contravariant 2-functor $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

$$(i) \ 1^\circ = 1 \quad (ii) \ (RS)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



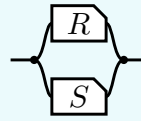
(3.1a)

R° is DEFINED as the bending of $(R \otimes 1) \triangleright \circ$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\circ \triangleleft$ bends the $\mathbf{a}$ strand down and the $\triangleright \circ$ brings it back up, while the $\mathbf{b}$ strand rides through untouched. Nothing else is spent below.

§4. $\cap$ is a commutative idempotent monoid on every hom-set

(4a)

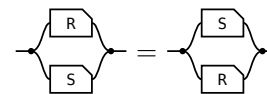
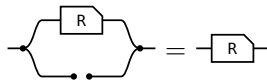
The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \langle \downarrow (R \otimes S) \rangle$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.



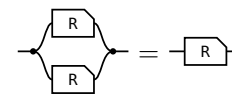
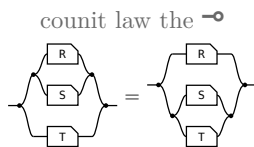
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $\top = \text{---} \bullet \bullet \text{---}$ (the paper's Lemma 4.11).

(4b)



unit: one half of $\top$ per end — the merge's unit law absorbs the $\bullet$, the copy's counit law the $\bullet$ **commutative:** σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

And one law relating $\cap$ to composition, which is **not** an equation:

(4c)

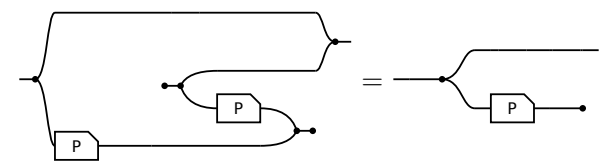
semi-distributivity, and what supplies it	picture
$R \cap (S \cap T) \leq R \cap S \cap T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F(R \cap S) = F R \cap F S$.	

§5. Domain and range

(5a)

The **domain** $\text{dom}(R) \triangleq 1 \cap R R^\circ$ and the **range** $\text{ran}(R) \triangleq \text{dom}(R^\circ)$.

(5b)



$\triangleright$ lands the return leg back on the value $\triangleleft$ handed out,
so $R^\circ \triangleright$ cuts to $\circ$ — Frobenius

Running R and throwing the result away leaves only the fact that R could fire, and $\text{ran}(R)$ is the same picture with the box mirrored. In Rel both steps are $\{(a, a) : \exists b. a R b\}$.

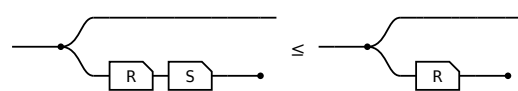
(5c)

$\text{dom}(R) \triangleq 1 \cap R R^\circ$
$\text{Coreflexive}(X) \implies \text{dom}(R) \subseteq X \iff R \subseteq X R$
$\text{dom}(RS) \subseteq \text{dom}(R)$
$\text{dom}(R \cap S) = 1 \cap S R^\circ$
$R \text{ entire} \iff \text{Entire}(R) \triangleq \text{dom}(R) = 1$
$R \text{ simple} \iff \text{Simple}(R) \triangleq R^\circ R \subseteq 1$
$R \text{ a map} \iff \text{Map}(R) \triangleq \text{Entire}(R) \wedge \text{Simple}(R)$
$\text{Entire}(R) \wedge \text{Entire}(S) \implies \text{Entire}(RS)$
$\text{Simple}(f) \wedge \text{Simple}(g) \implies \text{Simple}(fg)$
$\text{Map}(f) \wedge \text{Map}(g) \implies \text{Map}(fg)$
$\text{Entire}(RS) \implies \text{Entire}(R)$

§5.1. Sliding the discard

$\text{dom}(RS) \subseteq \text{dom}(R)$, and a single glyph for **dom** would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

(5.1a)



$S \circ \circ \subseteq \circ$, the lax axiom for $\circ$ in the first section
— the discard slides back past S

Equality is S entire, which is the same picture read as $\text{dom}(R) = 1 \iff R \text{ entire}$.

§6. Maps

(6a)

surjective $1 \sqsubseteq R^\circ R$, that is, R° entire.	single valued $R^\circ R \sqsubseteq 1$
entire $1 \sqsubseteq RR^\circ$. With single valued , a map .	injective $RR^\circ \sqsubseteq 1$

(6b)

law	picture
$\text{Simple}(F) \implies F(R \cap S) = F R \cap F S$ F single valued	 one person who admires both = a at A, b at B

§8. / is all of

(8a)

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

$$x (S \setminus R) y \iff \forall p. p S x \rightarrow p R y$$

/compares images: $S(y) \subseteq R(x)$. \ compares preimages: $S^\circ(x) \subseteq R^\circ(y)$.

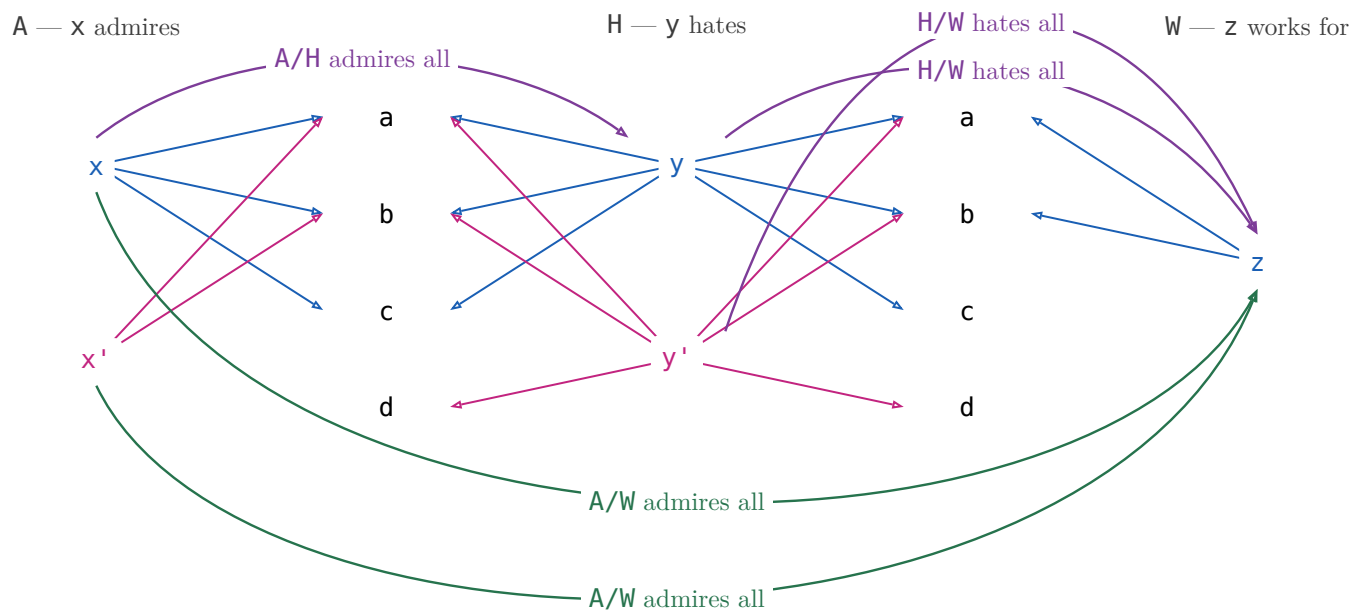
example: A admires, H hates, W works for

$x (A/H) y$ — x admires everyone y hates.

$x (H \setminus A) y$ — everyone who hates x admires y .

§8.1. $(R/S)(S/W) \subseteq R/W$

(8.1a)



(8.1b)

$$x (A/H) y \text{ — } x \text{ admires everyone } y \text{ hates}$$

$$y (H/W) z \text{ — } y \text{ hates everyone } z \text{ works for}$$

$$x (A/W) z \text{ — } x \text{ admires everyone } z \text{ works for}$$

$(A/H)(H/W)$ is a path: $x \rightarrow y \rightarrow z$, and that is all of it. A/W also holds of x' , who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \subseteq R/W$.

$X \in R/S \iff XS \in R$ X is any x -to- y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S , and R/S is the largest such.	
$X \in S \setminus R \iff SX \in R$ The mirror — divide on the left when x comes first.	
$(R/S)S \in R$ There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at $S=\emptyset$: R/S is everyone, $(R/S)S=\emptyset$.	
$S \setminus (S \setminus R) \in R$ The mirror.	
associate: $R/(S_1 S_2) = (R/S_2)/S_1$ A friend's enemy is two hops: divide by the far end first.	
$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	
maps: $f \setminus (R/S) = (fR)/S$ Rename x before or after dividing — the licence to write fR/S .	
$R/(fS) = (R/S) f^\circ$ Rename y : a map leaves a denominator as f° outside the box.	
$(R/S)(S/W) \in R/W$ Someone who admires all of a hate-set that already covers everyone z works for admires those people too.	
$1 \in R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
$(R/R)(R/R) = R/R$ R/R is the preorder admires at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \in R/R$ above it is an equality.	
$(R/R)R = R$ Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
$R/1 = R$ Dividing by 1 : p 's set is just $\{p\}$, so admiring all of it is admiring p .	
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Admiring a combined hate-set is admiring each set in full.	
$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: η , υ , $\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

§9. $\frac{R}{S}$

(9a)

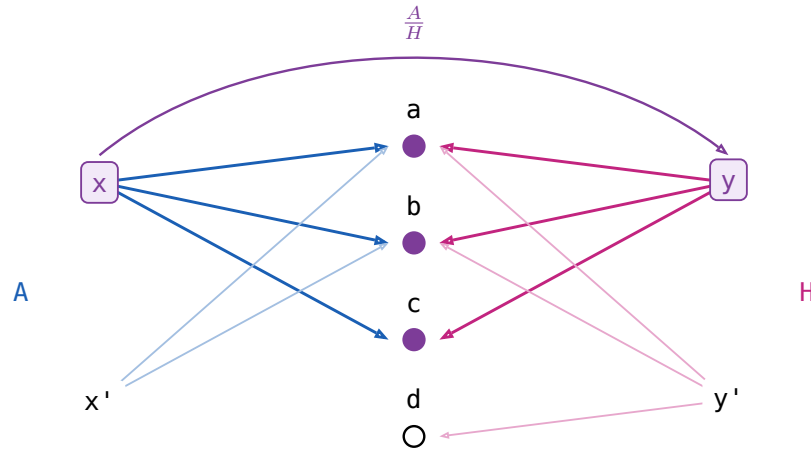
$R/\frac{S}{S} \triangleq (R/S) \cap (S/R)^\circ$ what R sends x to and nothing else is what S sends y to. In Rel x and y has the same image: $\forall p. (x R p \iff y S p)$

(9b)

$x (A/H) y$ — x admires everyone y hates
 $x ((H/A)^\circ) y$ — x admires only people y hates
 $x ((A/H) \cap (H/A)^\circ) y$ — x admires only and all whom y hates

x admires exactly whom y hates: $/$ is **all of**, $\frac{R}{S}$ is **only all of**.

(9c)



(9d)

law	the reading
$X \in R/\frac{S}{S} \iff X S \in R \wedge X^\circ R \in S$ X sits inside the matching exactly when it carries S back inside R and, reversed, carries R back inside S .	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
$(R/\frac{S}{S})^\circ = S/\frac{R}{R}$ reversing the matching swaps the two relations.	Matching is symmetric.
$(R/\frac{S}{S})(S/\frac{W}{W}) \in R/\frac{W}{W}$ a match from R to S followed by one from S to W is a match from R to W .	And transitive.
$(R/\frac{S}{S}) S \in R$ following the matching by S lands inside R .	$(\exists y. x(\frac{R}{S})y \wedge y S p) \rightarrow x R p$ x only admires whom y hates $\frac{R}{S} S = \text{Dom}(\frac{R}{S}) R$
$(R/\frac{S}{S}) R = R$ matching R against itself and then following R gives R back.	$(\exists y. x(\frac{R}{R})y \wedge y R p) \iff x R p$ x and y admire the same people $y=x$ always qualifies ($1 \in R \circ R$ below)
$1 \in R/\frac{S}{S} R$ every x matches itself.	$x(\frac{R}{R})y$ if x and y admires the same peoples.
$(R/\frac{S}{S})(R/\frac{S}{S}) = R/\frac{S}{S}$ matching twice matches no more pairs than matching once.	So the relation admires the same people is an equivalence relation.
$\text{Symmetric}(X) \implies X \in R/\frac{S}{S} R \iff X R \in R$ a symmetric X sits inside the matching exactly when following it by R adds nothing to R .	The largest symmetric arrow that leaves R alone.
$\text{simplePart}(R) \triangleq R/\frac{S}{S} 1$ matching R against the identity keeps the x whose image is a single point.	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R/1 = R$.
$\text{dom}(R/\frac{S}{S}) = 1 \cap (R/S)(S/R)$ the x that match something are those the two one-sided divisions take round the loop back to x .	Its domain is the domain of simplicity of R .

§10. Freyd's Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows, $\exists$ **epsilon**, subject to

$$\begin{aligned} \exists_R \square &= R\square, \quad \exists_R = \exists_R \square \\ 1 &\sqsubseteq (R/\exists_R)(\exists_R/R) & \exists_R \text{ is } \mathbf{thick} \\ \exists_R &= 1 & \exists_R \text{ is } \mathbf{straight} \end{aligned}$$

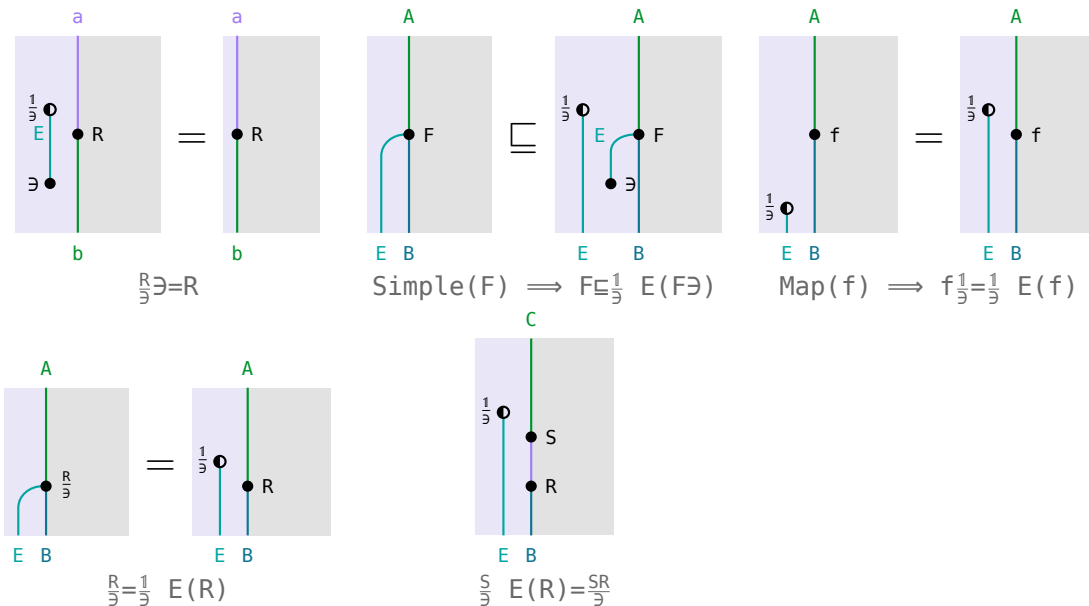
$R\square$ is R 's target, an identity arrow. For $R : A \rightarrow B$ write $\exists : EB \rightarrow B$, dropping the subscript.

$$\in \triangleq \exists^\circ : A \rightarrow EA$$

(10b)

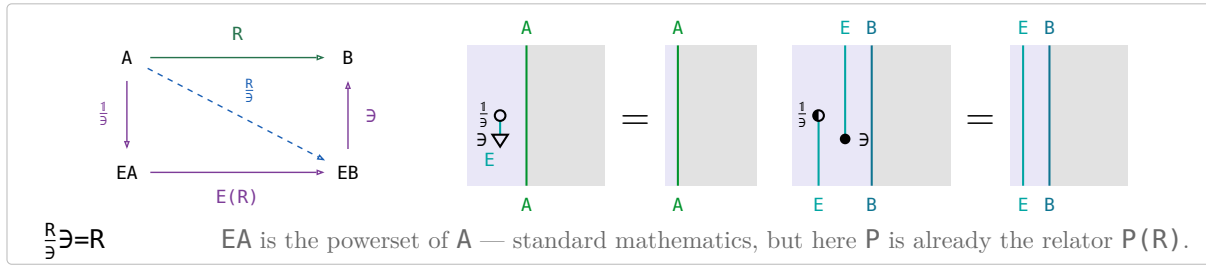
law	the reading
1 $\exists_R \square = R\square, \quad \exists_R = \exists_R \square$	One $\exists$ per object, not per arrow.
2 $\exists$ is thick	Comprehension: every x has a set of exactly the people x admires.
3 $\frac{R}{\exists} : A \rightarrow EB$, for $R : A \rightarrow B$	convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$
4 $\text{Map}(\frac{R}{\exists})$	
5 $\frac{R}{\exists} \exists = R$	reading the chosen set back through $\exists$ returns the relation.
6 $\text{Simple}(F) \Rightarrow F \sqsubseteq \frac{F\exists}{\exists}$	A partial choice of sets is inside the total one.
7 the singleton map is monic: $\frac{1}{\exists}$ $(\frac{1}{\exists})^\circ \sqsubseteq 1$ two points with the same one-person set are the same point.	The one-person set.
8 $\frac{\exists}{\exists} = 1$	Make the set of a set, then read it back one level down.
9 fusion: $\text{Map}(f) \Rightarrow \frac{fR}{\exists} = f \frac{R}{\exists}$	Naturality of the unit, $\text{Map}(f) \Rightarrow f \frac{1}{\exists} = \frac{1}{\exists} E(f)$. renaming a point and then taking its one-person set is taking the set and renaming inside it.
10 $\text{Map}(f) \wedge 1\square = \exists\square \Rightarrow \frac{f}{\exists} = f \frac{1}{\exists}$	Rename first or take singletons first — the fusion row above at $R=1$.
11 $R\square = \exists\square \Rightarrow R/\exists S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$	x and y match when R sends x and S sends y to the same set.
12 $E(R) \triangleq \frac{\exists R}{\exists}$	$E(R) : EA \rightarrow EB$, image of a set of A
13 $\frac{R}{\exists} = \frac{1}{\exists} E(R)$	$\frac{1}{\exists} : x \mapsto \{x\}$
14 $\frac{S}{\exists} E(R) = \frac{SR}{\exists}$	absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{SR}{\exists}$, §10.3
15 $\exists \triangleq \exists/\exists$	$xs \supseteq ys \iff \forall a. ys \exists a \rightarrow xs \exists a$
16 $\sqsubseteq \triangleq \exists^\circ$	$xs \sqsubseteq ys \iff ys \supseteq xs$

(10c)



§10.1. $\mathbf{i} \dashv \mathbf{E}$ Power Allegory defined as adjunction

(10.1a)



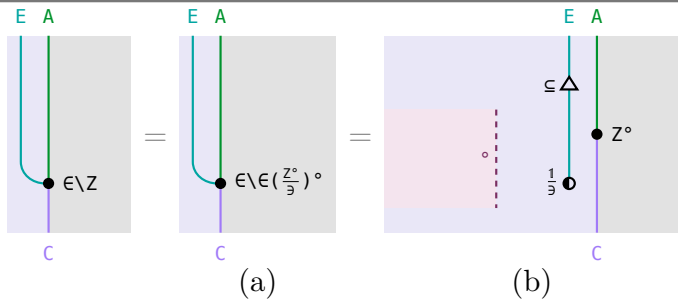
$\mathbf{i}$ is the inclusion $\mathbf{Map}(\mathcal{A}) \rightarrow \mathcal{A}$, doing nothing, so $\mathbf{E}$ is both the functor and the monad $\mathbf{iE}$.

§10.2. $\mathbf{E} \setminus$ as a composite

$$\mathbf{E} \setminus \mathbf{Z} = \mathbf{E} \left(\frac{\mathbf{Z}^\circ}{3} \right)^\circ$$

(10.2a)

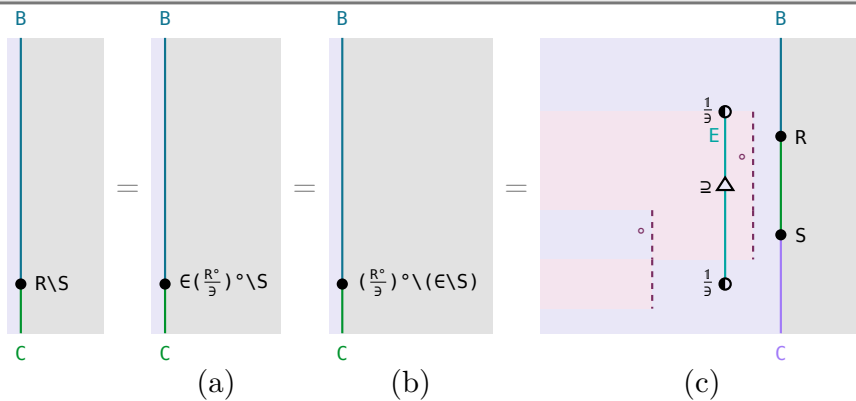
$\mathbf{x} \mathbf{s}$ is related to $\mathbf{c}$ by $\mathbf{E} \setminus \mathbf{Z}$ exactly when $\mathbf{x} \mathbf{s} \subseteq \mathbf{E} \left(\frac{\mathbf{Z}^\circ}{3} \right)^\circ (\mathbf{c})$, the set of every $\mathbf{a}$ with $\mathbf{a} \mathbf{Z} \mathbf{c}$



$$\mathbf{R} \setminus \mathbf{S} = \frac{\mathbf{R}^\circ}{3} \subseteq \left(\frac{\mathbf{S}^\circ}{3} \right)^\circ$$

(10.2b)

$\mathbf{b}$ is related to $\mathbf{c}$ by $\mathbf{R} \setminus \mathbf{S}$ exactly when $\mathbf{E} \left(\frac{\mathbf{R}^\circ}{3} \right)^\circ (\mathbf{b}) \subseteq \mathbf{E} \left(\frac{\mathbf{S}^\circ}{3} \right)^\circ (\mathbf{c})$, the $\mathbf{R}$ -preimage of $\mathbf{b}$ inside the $\mathbf{S}$ -preimage of $\mathbf{c}$



§10.3. $\mathcal{A} \cong \mathbf{Kleisli}(E)$

(10.3a)

$E(R) \triangleq \frac{\exists R}{\exists}$, $\frac{1}{\exists} : A \rightarrow EA$, $\text{union} = E(\exists) : E(EA) \rightarrow EA$

$f \diamond g \triangleq f E(g) \text{union}$, for $f : A \rightarrow EB$ and $g : B \rightarrow EC$

the monad is on $\mathbf{Map}(\mathcal{A})$, not on the allegory: E is a relator on all relations, but $\frac{1}{\exists}$, union and $f E(g) \text{union}$ are maps, and the Kleisli construction happens where they live.

(10.3b)

$$\begin{array}{c}
 \boxed{\frac{S}{\exists} \diamond \frac{R}{\exists}} \xrightarrow{\text{definition of } \diamond, \text{union} = E(\exists)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\right) E(\exists)} \xrightarrow{E \text{ a functor, } E(X)E(Y) = E(XY)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\exists\right)} \\
 \xrightarrow{(10b), \frac{R}{\exists}\exists = R} \boxed{\frac{S}{\exists} E(R)} \xrightarrow{(10b), \text{absorption}} \boxed{\frac{SR}{\exists}}
 \end{array}$$

the first three steps only unfold $\diamond$ — E is a functor, so $E\left(\frac{R}{\exists}\right)\text{union} = E\left(\frac{R}{\exists}\exists\right) = E(R)$ and the union is gone.

Absorption, the row of (10b), is the whole law, and it is the functoriality of $\frac{1}{\exists}$.

1 goes to $\frac{1}{\exists}$, the Kleisli identity, by definition — so $\frac{1}{\exists}$ is an isomorphism of categories $\mathcal{A} \cong \mathbf{Kleisli}(E)$, and §10.1's $i \dashv E$ is the Kleisli adjunction. i is the identity on objects, so every object of the allegory is free.